

Name: _____

Date: _____

Standards: MA.5.AR.1.1; MA.5.AR.3.1-3.2

Answer each question. Show your work when asked.

This check covers the lessons listed above.

1 Unit Rate

A car travels 350 miles in 5 hours.

Find the unit rate.

 $350 \div 5 = 70$ miles per hour. _____**2 Distance Formula**

A plane flies at 600 km/h for 2.5 hours.

How far does it travel?

 $600 \times 2.5 = 1,500$ km. _____**3 Find the Time**

A hiker walks at 4 km/h and needs to travel 18 km.

How long will the hike take?

 $18 \div 4 = 4.5$ hours. _____

4 Compare Rates

Printer A prints 40 pages in 5 minutes. Printer B prints 90 pages in 10 minutes.

Which printer is faster? Find each rate.

A: $40 \div 5 = 8$ pages per minute. B: $90 \div 10 = 9$ pages per minute. Printer B is faster.

5 Earnings

Leo earns \$9.50 per hour. He works 8 hours.

How much does he earn?

$\$9.50 \times 8 = \76 .

6 Better Buy

6 cans of soup cost \$4.80. A pack of 10 cans costs \$7.50.

Which is the better deal per can?

6 cans: $\$4.80 \div 6 = \$0.80/\text{can}$. 10 cans: $\$7.50 \div 10 = \$0.75/\text{can}$. The pack of 10 is better value.

7 Find the Speed

A bus covers 72 km in 1.5 hours.

What is the bus's average speed?

$$72 \div 1.5 = 48 \text{ km/h.}$$

8 Two Workers

Mira can pack 30 boxes per hour. Atlas can pack 45 boxes per hour.

Working together, how many boxes can they pack in 3 hours?

$$\text{Combined rate} = 75 \text{ boxes/hour. } 75 \times 3 = 225 \text{ boxes.}$$

9 Multi-Step

A car travels 60 km/h for 1 hour 30 minutes, then 80 km/h for 30 minutes.

What is the total distance?

$$60 \times 1.5 = 90 \text{ km; } 80 \times 0.5 = 40 \text{ km. Total} = 130 \text{ km.}$$

10 Is Mira Correct?

Mira says that if you double the speed, you halve the time to travel the same distance.

Is she correct? Give a numerical example.

Yes. Example: 60 km at 30 km/h takes 2 hours. At 60 km/h it takes 1 hour. Doubling the speed halves the time. _